

AERO THERMODYNAMICS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE05	ESC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100
UNIT-I	BASIC CONCEPT AND FIRST LAW OF THERMODYNAMICS							
Basic Concepts: Macroscopic and Microscopic viewpoints, Quasi static Process, various flow and non-flow process, energy in State and in Transition, Work and Heat, Point and Path function. Zeroth Law of Thermodynamics, First law of Thermodynamics, First law applied to a Process, applied to a flow system.								
UNIT-II	SECOND LAW OF THERMODYNAMICS AND ENTROPY PRINCIPLES							
LIMITATIONS OF THE FIRST LAW: Thermal Reservoir, Heat Engine, Heat pump Refrigerator. Parameters of performance, Second Law of Thermodynamics-Kelvin Planck and Clausius Statements and their Equivalence / Corollaries, PMM of Second kind, Carnot's principle, Carnot cycle and its specialties, Clausius Inequality, Entropy, Principle of Entropy Increase, Elementary Treatment of the Third Law of Thermodynamics.								
UNIT-III	VAPOUR AND GAS POWER CYCLES							
VAPOUR POWER CYCLES: P-V-T surfaces, T-S and h-s diagrams, Mollier Charts, Phase transformations, Dryness Fraction. Various thermodynamic processes and energy transfer. Simple Rankine cycle. GAS POWER CYCLES: Otto, Diesel and Brayton cycles, Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis, comparison of Cycles. Basics of Jet propulsion.								
UNIT-IV	GAS TURBINE ENGINE COMPRESSORS							
Axial Flow Compressors: Elementary theory of axial flow compressor, Velocity triangles, Degree of reaction. Flow through an Axial flow Compressor stage; Losses in axial flow compressor stage; Compressor blade design, Stage losses, Characteristics of Blade - single and multi-stage Axial compressor.								
UNIT-V	GAS TURBINE ENGINE TURBINES							
Turbines: Introduction; Turbine stage; Turbine Blade 2-D (cascade) analysis Work Done; Degree of Reaction; Losses and Efficiency; Flow Passage; Subsonic, transonic and supersonic turbines, Multi-staging of Turbine, Turbine Cooling.								
Text Books:								
1. Engineering Thermodynamics, P.K. Nag, 6 th Edition, Mc Graw Hill Education. 2. Gas Turbines, V Ganesan, 3 rd Edition, Mc Graw Hill Education (India) Private Limited								
Reference Books:								
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2 nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Gas Turbine Theory, H.I.H Saravanamuttoo, G.F.C Rogers and H. Colen, 7 th Edition, Pearson								

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Apply first law of thermodynamics applied to engines
2. Evaluate the performance parameters of energy transfers
3. Evaluate the properties of gases during phase changes and differentiate the processes of various power cycles
4. Analyze the characteristics of axial flow compressor
5. Analyze the characteristics of turbine